

2024



Paytm Realizes 35% Savings and Up to 30% Application Performance Improvement using AWS Graviton

Learn how Paytm—India’s leading payment and financial services platform—optimized its costs and improved performance by migrating to AWS Graviton.

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30–35% cost savings

on Amazon EMR workloads

20–30% throughput improvements

of MySQL workloads

60%

of EC2 instances migrated to AWS Graviton



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Up to 47%

estimated decrease in workload carbon intensity (carbon emissions per transaction)



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Overview

[Paytm Mobile Solutions](#) (Paytm) has a massive infrastructure and scales to accommodate millions of active users, so keeping costs down is vital. In 2021, Paytm decided to optimize its compute costs on AWS.

Since 2013, the company has deployed its applications on various services in Amazon Web Services (AWS), including [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. So, when AWS announced the Mumbai Region release of Amazon EC2 instances powered by [AWS Graviton processors](#)—a family of processors designed to deliver the best price performance for cloud workloads running on Amazon EC2—the company chose to migrate its data lake, MySQL workloads, AWS managed services (including RDS, Aurora, Opensearch, ElastiCache, etc), and payment gateway applications. Even though Paytm initially focused on cost optimization in its migration, the company also realized performance and sustainability improvements.



Opportunity | Using AWS Graviton to Optimize Costs for Paytm

One97 Communications Limited (OCL) that owns the brand Paytm is India's leading payments and financial services distribution company and the pioneer of QR, Soundbox, and mobile payments. In its mission to bring 500 million unserved and underserved Indians to the mainstream economy, Paytm runs multiple businesses across payments, loan distribution, wealth management, insurance distribution, and online travel and ticketing etc.

AWS Graviton-based Amazon EC2 instances in AWS Mumbai region (ap-south-1) are priced almost 50 percent lower than comparable instances. Paytm migrated to AWS Graviton to optimize EC2 costs. "When AWS announced the Mumbai Region release of Amazon EC2 instances powered by AWS Graviton processors in the Mumbai Region, we were one of the first companies to do a pilot migration," says Paytm. On Graviton based instances, Paytm saw better price and performance for their applic



executing a proof of concept and saw the advantages of using AWS Graviton in multiple applications throughout the company.

Solution | Saving Up to 35 % on Compute Costs Using AWS Graviton

Paytm first migrated its MySQL workloads in mid-2021, including a self-managed MySQL cluster with over 160 nodes in production, to AWS Graviton-based instances. To do this, Paytm had to first update its MySQL database to a minimum software version recommended for AWS Graviton processors, and because of the criticality of the databases as well as dependencies between workloads, this update was a complex undertaking. “The AWS teams supported us very well, especially considering the scale and speed at which Paytm works,” says Parakh Aggarwal, Director of Cloud Excellence and FinOps. Paytm not only realized a 50% reduction in prices for EC2 instances, but also saw 20–30 percent performance improvements in its MySQL workloads (24% in Transaction Per Second and 30% in Queries Per Second) compared with its previous solution. This workload migration helped Paytm realize the potential savings from Graviton adoption, which laid the foundation for a systematic and program driven expansion of Graviton usage across multiple workloads in Paytm.

For its payment gateway migration, Paytm migrated 80 percent of its more than 1,000 Amazon EC2 instances to AWS Graviton-based instances by December 2023. Nearly 70 microservices were successfully migrated for the payment gateway alone. After migrating this portion of the payment gateway infrastructure, Paytm achieved an overall cost reduction of up to 35% of their Amazon EC2 spends. Additionally, Paytm improved the performance of its payment gateway services for both end users and merchants after migration. The payment gateway’s throughput improved by 47% in over a 6 months period from Mar to Sep 2023, when the Graviton adoption for this workload increased from 8% to 30%. Due to better performance of the workload running on Graviton, Paytm operated the workload more efficiently, leading to an estimated decrease in workload carbon intensity (workload carbon emissions per transaction) up to 47%.

For its data lake, Paytm had been using [Amazon EMR](#)—a cloud native big data solution for petabyte-scale data processing, interactive analytics, and machine learning—for around 65 percent of its analytics workloads. Paytm began the migration of these workloads to AWS Graviton-based instances in 2022 and achieved 30–35 percent cost savings. By the end of 2023, 80 percent of Paytm’s Amazon EMR workloads ran on AWS Graviton, resulting in better performance. “One benefit of migrating AWS managed workloads to AWS Graviton-based instances is that the functional compatibility and performance of these workloads is managed by AWS and, as a result, we have fewer complexities to manage,” says Aggarwal.



Outcome | Prioritizing Performance as Migration to AWS Continues

Paytm continues to migrate its massive infrastructure to AWS Graviton-based Amazon EC2 instances (60% of EC2 migrated to Graviton as of Jan-2024) and expects to have 70 percent of its infrastructure migrated by mid-2024. Paytm plans to migrate all its 200-plus services to AWS Graviton eventually. Now that Paytm has optimized costs, the next area of focus will be on additionally improving performance of all workloads and rightsizing EC2 instances to reduce costs further.

“Migrating to AWS Graviton-based instances is a long-term investment and will be beneficial in terms of total cost of ownership,” says Aggarwal.

About Paytm

One97 Communications Limited (OCL) that owns the brand Paytm is India’s leading payments and financial services distribution company and the pioneer of QR, Soundbox, and mobile payments.

AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 800 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

AWS Graviton Processor

AWS Graviton is a family of processors designed to deliver the best price performance for your cloud workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

[Learn more »](#)

Amazon EMR



Amazon EMR is the industry-leading cloud big data solution for petabyte-scale data processing, interactive analytics, and machine learning using open-source frameworks such as Apache Spark, Apache Hive, and Presto.

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